

RadonPy: From Automated Molecular Dynamics Simulations to Real-World AI Applications:

Transfer Learning & Scaling Law for Polymers

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Machine Learning for Polymer Science: TL;DR

Motivation: Predict polymer properties accurately to accelerate materials discovery.

Problem: Each lab often has only small, task-specific datasets, making it difficult to train reliable predictive models from scratch.

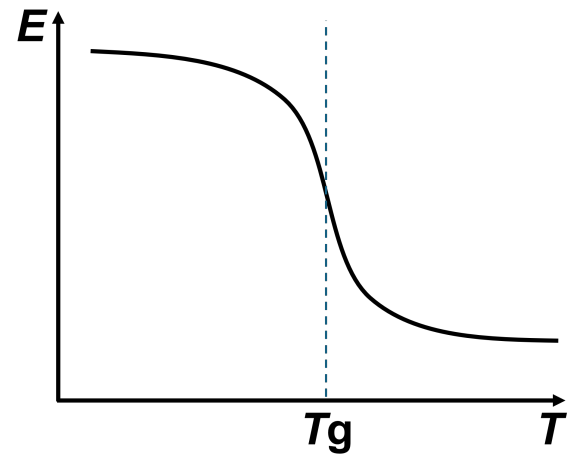
Solution: Pretrain on large-scale polymer data, then fine-tune on your own task.

Why polymer ML is hard

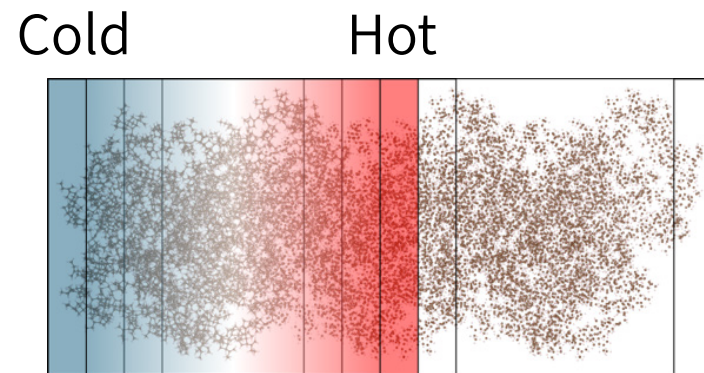
Many highly specialized tasks

- Polymer science has diverse tasks

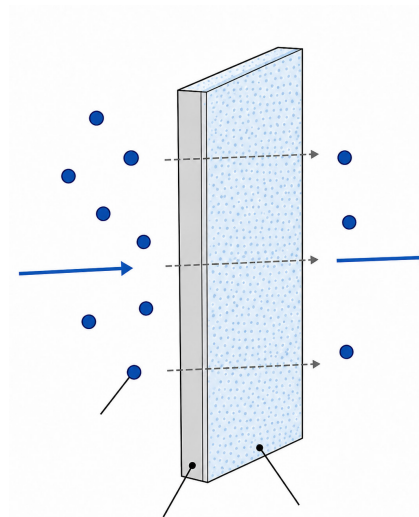
Glass transition temp.



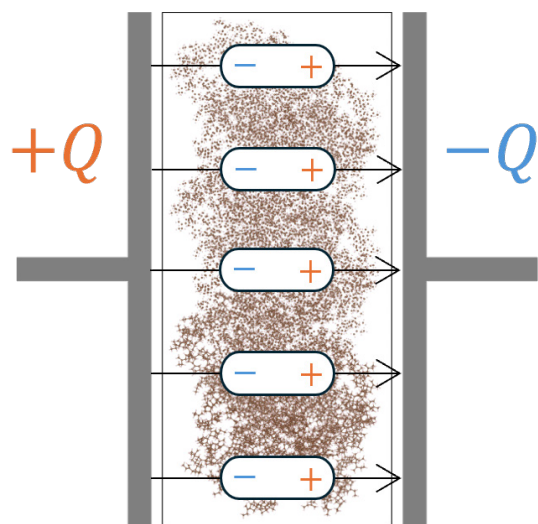
Thermal conductivity



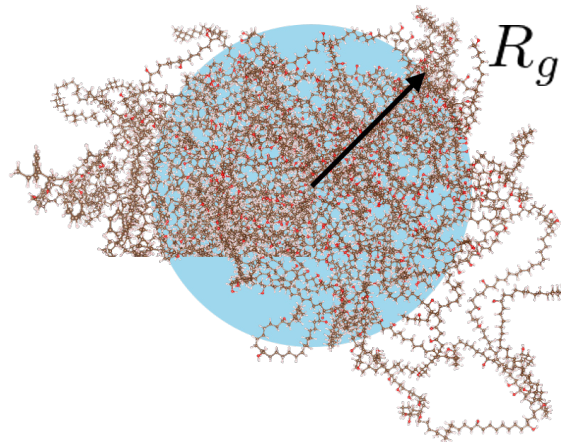
Gas permeability



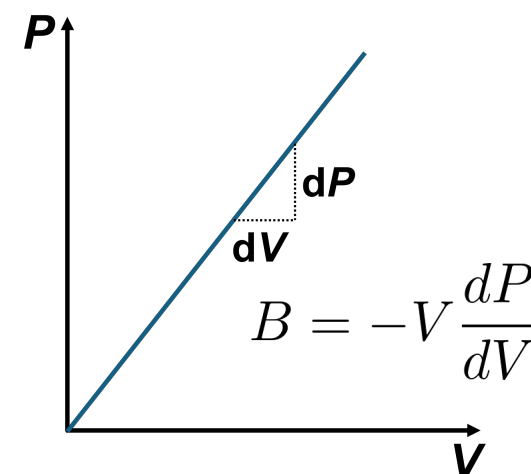
Dielectric constant



Radius of gyration



Bulk modulus



Each group has its own task-specific data

- Tasks are specialized
- Data are often small, fragmented, and difficult to share.

Lab 1



Polymer	Tg [K]	Density [g/cm ³]
P1	456	0.87
P2	223	0.99
P3	365	0.78

Lab 2



Polymer	Water absorption	Tensile modulus[GPa]
P4	0.6	1.9
P5	4.2	0.69
P6	7.9	1.4

Company 1



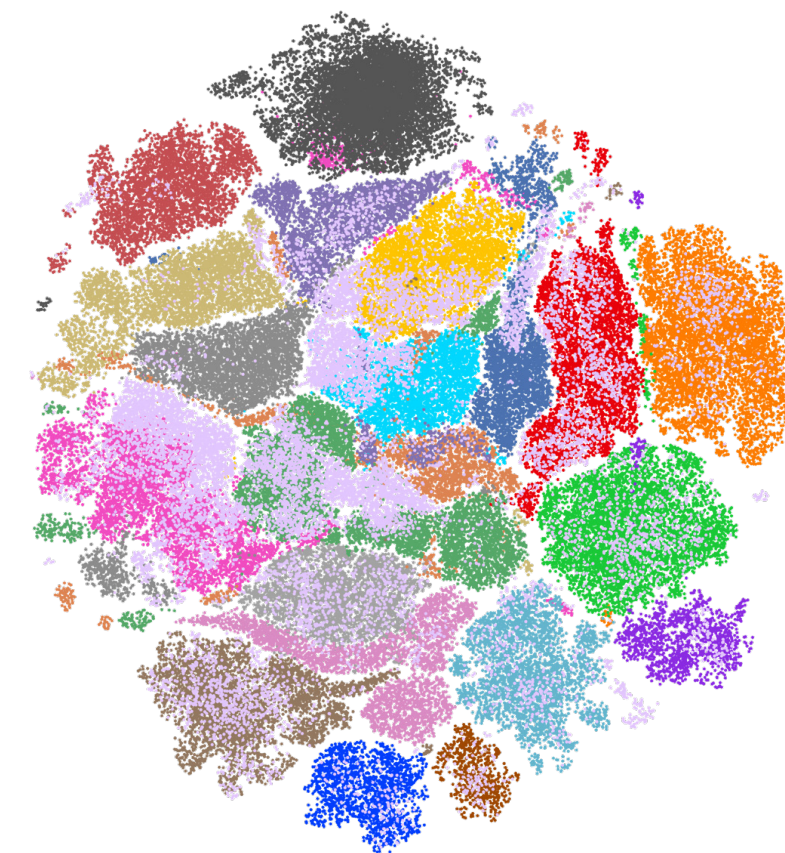
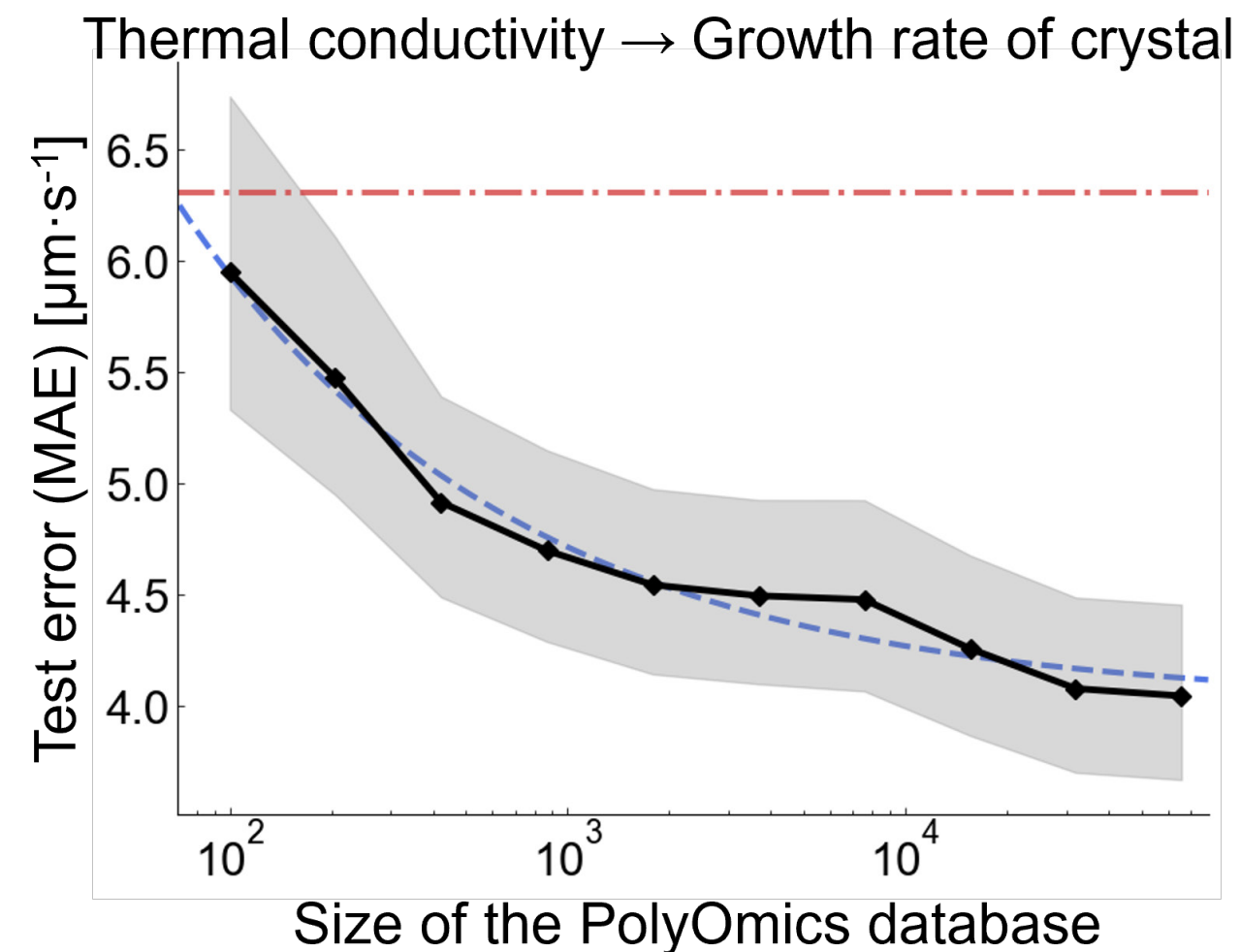
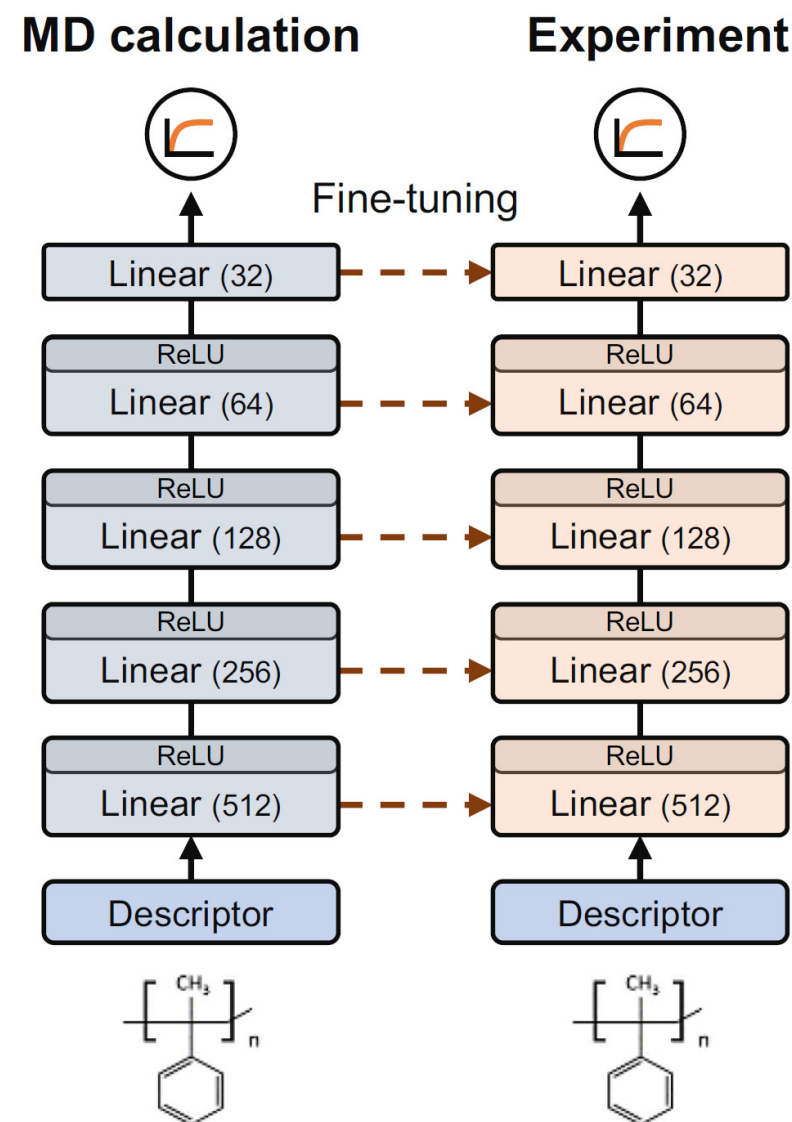
Polymer	Tm [K]	Solubility
P1	665	20.1
P4	453	16.8
P7	876	19.3

Polymer ML is not one task — it is many specialized tasks with many local datasets.

Solution: scaling law and foundation-scale polymer data

Scaling law and foundation-scale polymer data

- Minami et al. showed a scaling law for polymer property prediction.
- Prediction error decreases as the pretraining dataset grows.
- **PolyOmics** provides a large-scale polymer dataset for transfer learning.



- 130k polymers
- 5 million data points
- 21 polymer classes
- 62 properties
 - T_g
 - Thermal conductivity
 - R_g
 - Bulk modulus
 - etc.